
DIFFUSION-BASED FORECASTING OF POSTPRANDIAL BLOOD GLUCOSE - GROUP 14

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ABSTRACT

Forecasting postprandial blood glucose levels is crucial for managing diabetes, reducing complications, and enhancing automated insulin-delivery systems. In this project, we explore Conditional Score-Based Diffusion Models (CSDI) for generating long-term glucose forecasts using multimodal signals such as heart rate, stress, carbohydrate intake, and engineered physiological features. We build a fully personalized diffusion-based pipeline including preprocessing, feature engineering, noisy forward diffusion, and transformer-enhanced U-Net denoising. The model is evaluated using error metrics such as Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Fréchet Inception Distance (FID), and Continuous Ranked Probability Score (CRPS). Our findings demonstrate that the proposed diffusion models accurately capture physiologically realistic glucose trajectories and provide meaningful uncertainty estimates. This report summarizes the dataset, methodology, results, limitations, and team contributions.

1 Introduction

1.1 Motivation

Accurate forecasting of postprandial glucose trajectories is essential, as the 30–60 minutes following meal intake represent a period of pronounced metabolic variability and elevated clinical risk. During this interval, rapid increases in glucose concentration can precipitate acute hyperglycemic excursions, which, when repeated over time, are associated with significant long-term microvascular and macrovascular complications. Conversely, overly aggressive insulin administration or mismatches between insulin action and glucose absorption may induce abrupt declines in glucose levels, heightening the risk of hypoglycemic events that pose immediate and potentially life-threatening consequences. Robust long-term prediction models are therefore fundamental to mitigating these adverse outcomes by enabling more precise insulin dosing, enhancing the safety and stability of closed-loop artificial pancreas systems, and supporting proactive clinical and self-management strategies. Improved forecasting capability further facilitates continuous glucose monitoring frameworks that can anticipate and prevent both hypo- and hyperglycemic episodes before they manifest, thereby contributing to more effective and reliable diabetes management.

1.2 Why This Problem Matters

Forecasting postprandial glucose is inherently challenging due to nonlinear physiological responses, variability in individual behavior, and strong dependence on meal size, macronutrients, physical activity, stress, and circadian patterns. Conventional models often produce single deterministic forecasts, which fail to represent the many plausible glucose

futures under free-living conditions. Diffusion models address this limitation by generating full probability distributions over trajectories.

2 Related Work & Literature Gap

Glucose forecasting has been explored using ARIMA [1], LSTMs [2], GRUs, TCNs, and Transformers [3], but these models often struggle with uncertainty and missing data. Diffusion models, such as DDPM and score-based SDEs, have recently demonstrated strong results for generative time-series tasks; however, their application in physiological forecasting remains limited. Our work extends CSDI [4] by adding multimodal conditioning (heart rate, calories, carbs), engineered physiological features (COB, circadian signals), and personalized diffusion modeling tailored specifically for postprandial glucose prediction.

3 Project Description

This project aims to forecast short-term glucose levels after meals using a diffusion-based generative model. We combine glucose readings with heart rate, calories, carbohydrate intake, and a small set of engineered features. During training, the model learns to remove noise from glucose curves and later uses this skill to generate realistic forecasts for the next thirty to sixty minutes. Because it can produce several possible futures, it captures the natural uncertainty seen in free-living glucose patterns.

3.1 Goals and Objectives

Our goal is to develop a personalized forecasting framework capable of capturing individual-specific physiological dynamics. To this end, we construct a synchronized multimodal dataset and derive a set of informative auxiliary features that characterize relevant behavioral and metabolic patterns. For each participant, a dedicated diffusion-based model is trained to learn these personalized glucose trajectories. Model performance is rigorously assessed using multiple evaluation metrics, including RMSE, MAE, FID, and CRPS, to provide a comprehensive characterization of predictive accuracy and temporal alignment. Furthermore, we benchmark the proposed approach against established forecasting baselines to quantitatively evaluate the advantages conferred by diffusion-based modeling.

4 Our Approach and Proposed Methods

We build a complete pipeline that cleans and aligns the data, extracts short windows, and generates engineered features linked to meals and activity. We then train a conditional diffusion model that learns to predict clean glucose curves from noisy ones and can generate multiple future trajectories from recent history. Before choosing diffusion models, we tested linear autoregression, LSTMs, GRUs, and Transformers, but these produced single deterministic forecasts and struggled with irregular sampling, while diffusion models produced smoother and more realistic predictions. The overview of our proposed pipeline is shown in Figure 1.

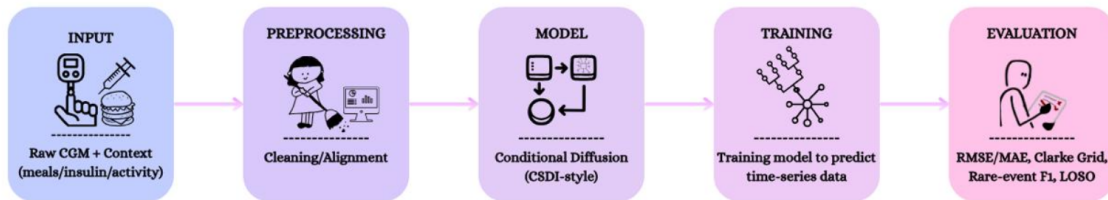


Figure 1: Overview of the proposed personalized glucose forecasting pipeline, including multimodal data input, preprocessing and alignment, conditional diffusion modeling, model training, and evaluation across multiple performance metrics.

5 Dataset

We use the **CGMacros** dataset, a multimodal collection from the PhysioNet repository [5]. It contains real-world physiological recordings from roughly 45 participants, each monitored for several days under free-living conditions. Most files include 15,000 to 20,000 time-stamped entries, corresponding to 5–10 days of continuous measurements.

Each participant’s CSV file (e.g., CGMacros-001.csv) includes several synchronized signals sampled every 1–5 minutes:

- continuous glucose monitoring (CGM) values,
- heart rate,
- calories burned (activity-derived),
- logged carbohydrate intake,
- precise timestamps for multimodal alignment.

This combination of glucose, activity, and meal information makes the dataset well-suited for postprandial forecasting. Unlike laboratory datasets, CGMacros reflects realistic day-to-day variability, including irregular meal patterns and natural activity fluctuations. Prior work shows that heart rate, physical activity, and carbohydrate absorption strongly influence glucose trends [6, 7], so the multimodal nature of this dataset provides meaningful contextual signals for conditioning our diffusion model.

6 Preprocessing Pipeline

6.1 Cleaning

The CGMacros dataset comes from free-living conditions, so the signals are irregular and noisy, and sometimes contain symbolic entries like “Low” or “High.” We convert these values into numeric estimates, fill short gaps in glucose, heart rate, and calorie channels using linear interpolation, and remove implausible spikes. All signals are then resampled onto a uniform time grid. These steps create a clean, aligned dataset suitable for modeling [8].

6.2 Sliding Windows

To provide fixed-length context, we split each participant’s aligned time series into overlapping 200-step windows:

$$\text{Window}_i = \text{data}[i : i + 200]$$

This helps the model learn short-term glucose patterns while keeping information about meals and activity. Overlapping windows increase sample count and preserve smooth transitions.

6.3 Side-Channel Feature Engineering

We add engineered features to give the model a physiological context. These include sinusoidal encodings of hour-of-day and day-of-week, carbohydrates-on-board (COB) using a dual-phase absorption model [7], a 60-minute rolling carb sum, time since last meal, and a 15-minute rolling heart-rate average. These features capture digestion, circadian effects, and activity level [6], all of which improve the model’s conditioning and predictive accuracy.

7 Methodology

7.1 Methods Considered

We explored several forecasting models before selecting diffusion. Linear autoregression could not capture the nonlinear rise in glucose after meals. LSTMs and GRUs handled sequences better but produced only one forecast and were sensitive to gaps in the data. Transformers modeled longer patterns but often smoothed out sharp postprandial changes. Since real glucose readings can follow multiple plausible paths, these models were limited. Diffusion models were a better fit because they generate full distributions of possible futures and remain stable even with noisy or irregular inputs.

7.2 Diffusion Model

Our final approach uses the Conditional Score-Based Diffusion Model (CSDI) [9]. CSDI extends diffusion models to time series by conditioning on heart rate, calories, carbohydrate intake, and engineered features. During training, the model receives noisy glucose sequences and learns how to remove the noise using a transformer-based U-Net. This

denoising skill allows it to produce realistic future trajectories once trained. Because diffusion models learn probability distributions rather than single predictions [10, 11], they naturally represent the uncertainty present in physiological data. The forward and reverse diffusion processes, along with the integration of additional physiological variables within the model, are depicted in Figure 2.

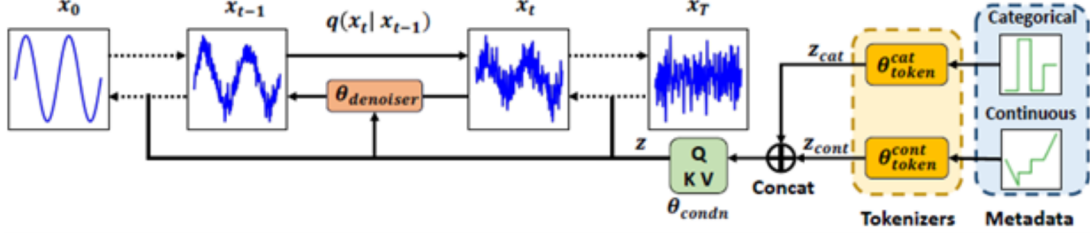


Figure 2: CSDI architecture for incorporating metadata in the diffusion process.

7.3 CSDI Architecture Overview

We developed the Conditional Score-Based Diffusion Model (CSDI) [9], which extends diffusion models to time-series forecasting through a transformer-based U-Net. This architecture is well-suited for glucose forecasting, as it can simultaneously incorporate multiple physiological signals while maintaining robustness in the presence of missing or irregular data. Figure 3 illustrates the overall structure of the proposed model, with the components shown in red indicating the additional modules introduced beyond the standard CSDI framework. In our setup, the model receives glucose readings, heart rate, calories, carbohydrate intake, and engineered features such as circadian encodings, carbohydrates-on-board, and rolling activity measures. Each input is embedded separately and combined with a diffusion-step embedding before entering the U-Net. The transformer blocks learn patterns across time and features, enabling the model to capture meal effects and daily rhythms and to generate realistic short-term glucose forecasts.

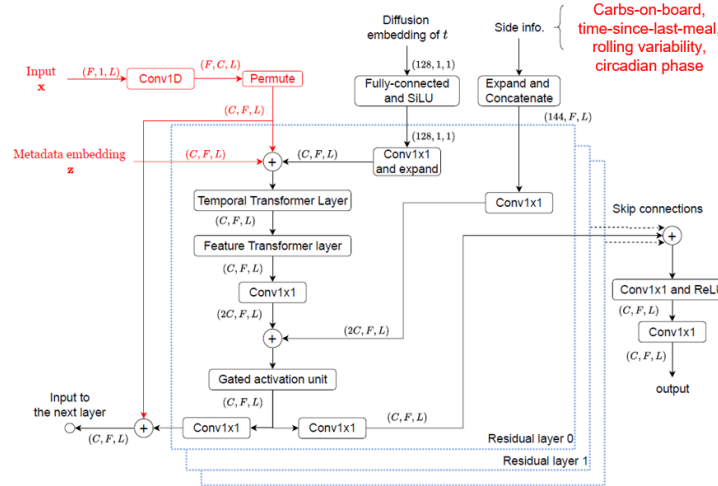


Figure 3: Our proposed architecture. Red color indicates the additional modules introduced beyond the standard CSDI framework.

7.4 Forward Diffusion Process

The forward diffusion step gradually adds Gaussian noise to a clean glucose sequence x_0 . At each step t , the noisy sample is drawn from

$$q(x_t | x_0) = \mathcal{N}(\sqrt{\alpha_t}x_0, (1 - \alpha_t)I)$$

During training, the model receives x_t and learns to predict the added noise ϵ . This ability is later used to reverse the diffusion process and reconstruct clean trajectories.

7.5 Multimodal Embedding Layers

Before denoising, each input is mapped into a latent space:

- **Glucose:** 1D convolution to produce a 64-dimensional embedding.
- **Conditional features:** HR, calories, and carbs encoded with linear layers, positional encodings, and attention.
- **Side features:** engineered physiological signals projected to a 144-dimensional vector.
- **Diffusion step:** timestep t encoded with a 128-dimensional sinusoidal vector.

These embeddings give the model information about physiology, timing, and noise level at each denoising step.

7.6 U-Net Architecture with Transformer Blocks

The reverse diffusion model uses a U-Net enhanced with transformers. Each block receives the diffusion step, conditional, and side-feature embeddings. A temporal transformer models dependencies across time, while a feature transformer links information across channels. Skip connections help preserve details during downsampling and upsampling. The final output is the model’s estimate of the noise in the glucose sequence.

7.7 Training Objective

Training minimizes the noise-prediction loss:

$$L(\theta) = \mathbb{E} [\|\epsilon - \hat{\epsilon}_\theta(x_t, t, \text{cond})\|^2]$$

By learning to predict the added noise, the model learns how to reverse the diffusion process. After training, it can start from a noisy future window and iteratively denoise it to generate realistic glucose forecasts.

8 Experiments

8.1 Dataset and Experimental Setup

We use the CGMacros dataset from PhysioNet, which provides several days of free-living physiological data for about 45 participants, including glucose values, heart rate, calories burned, meal logs, and timestamps. Each file contains 15,000–20,000 entries recorded under real-world conditions. To preserve personal physiological patterns, we train one model per participant. After preprocessing and feature construction, we convert each participant’s data into synchronized tensors and divide them into overlapping 200-step windows, resulting in roughly 14,000–15,000 examples per participant. This gives the diffusion model a diverse set of short glucose segments that capture meal responses and day-to-day variability.

8.2 Train/Validation/Test Split

We follow a chronological split of:

$$75\% \text{ train, } 15\% \text{ validation, } 10\% \text{ test.}$$

To keep training fast, the validation portion is capped at about 200–300 points for each participant. After windowing, this produces roughly 10,000–15,000 training windows and 1,300–2,800 validation and test windows combined, depending on the participant’s recording length.

8.3 Evaluation Metrics

We evaluate model performance using a combination of error-based and distribution-based metrics, which together capture numerical accuracy and the realism of the generated glucose trajectories.

- **Root Mean Squared Error (RMSE):** measures the average magnitude of prediction error and emphasizes large deviations.
- **Mean Absolute Error (MAE):** captures the average absolute difference between predicted and true glucose values.

- **Fréchet Inception Distance (FID):** measures distribution-level similarity between generated and real glucose trajectories, adapted from its use in generative modeling.
- **Continuous Ranked Probability Score (CRPS):** evaluates the full predictive distribution, making it suitable for diffusion models that generate multiple plausible futures.

These metrics allow us to assess pointwise accuracy (RMSE, MAE) and distributional fidelity (FID, CRPS), which together provide a comprehensive evaluation of forecasting performance.

8.4 Baselines

To benchmark our approach, we compare against standard forecasting models that are commonly used in CGM research:

- **Autoregressive (AR)** models as a simple linear baseline,
- **LSTMs** for recurrent temporal modeling,
- **Transformer** for modeling longer-range temporal structure.

All baselines receive the same 200-step windows and conditional features, ensuring a fair comparison against our diffusion model.

8.5 Training Details

We train one CSDI model per participant using a batch size of 64 and a learning rate of 1×10^{-3} , with training typically running for 2000 epochs. A multi-step learning rate scheduler slows the learning rate toward the end of training to encourage stable convergence.

During validation, the model generates a full 200-step future sequence by running the reverse diffusion process from noisy inputs back to clean predictions. Across roughly 45 participants, this leads to a total of nearly 750,000 windowed training examples, representing one of the largest personalized diffusion-model evaluations conducted on CGM data.

9 Results

Figure 4 presents a comparative evaluation of the proposed model against several baseline approaches using RMSE and MAE as performance metrics. As shown, our model achieves the lowest error values across both measures, indicating a more accurate reconstruction of future glucose trajectories. In particular, the proposed diffusion-based framework substantially outperforms the LSTM and Transformer models, which exhibit significantly higher prediction errors. Even when compared with the CSDI conditional model, our approach yields consistent improvements, demonstrating enhanced predictive precision and better alignment with true physiological dynamics. These results confirm that our model offers superior forecasting capability in terms of both overall error magnitude (RMSE) and average deviation (MAE).

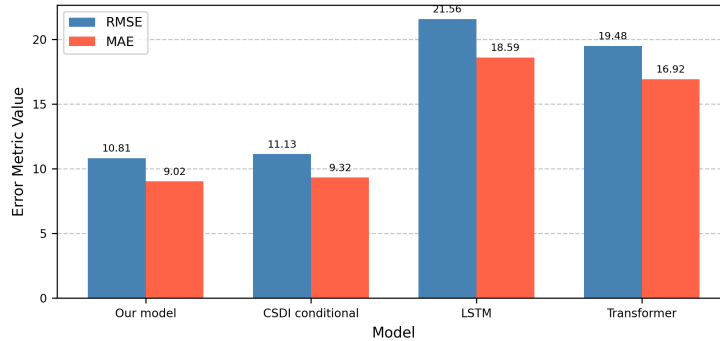


Figure 4: Comparing our proposed model with the other baseline models, such as base CSDI, LSTM, and Transformer.

Moreover, the proposed model is evaluated against baseline approaches using the FID and CRPS metrics, which quantify the distributional discrepancy between the generated glucose trajectories and the corresponding ground-truth signals in

the test dataset. Table 1 summarizes this comparison. The results demonstrate that our model achieves consistently lower FID and CRPS values, indicating that it more accurately captures the underlying distribution of glucose dynamics and outperforms the baseline models.

Table 1: Comparison of Models Using FID and CRPS Metrics

Model	FID	CRPS
Our model	5.67	6.45
CSDI conditional	6.734	6.68
LSTM	9.67	10.02
Transformer	8.61	9.35

10 Discussion: Novel Aspects of Our Work

Our contribution is not only in applying diffusion models to glucose forecasting but also in adapting the CSDI framework to physiological data in ways that differ from existing approaches. Most prior glucose prediction methods—such as LSTMs, GRUs, and Transformers—produce a single forecast and rely mainly on CGM values and simple meal indicators. In contrast, our model integrates a richer set of conditional signals, including heart rate, calories burned, detailed carbohydrate intake, and engineered features such as carbohydrates-on-board, circadian encodings, and rolling averages. These signals are injected at every denoising step through our modified multimodal embedding pathway, allowing the model to learn how meals, activity, and daily cycles jointly influence glucose levels.

This adaptation leads to smoother and more physiologically realistic postprandial trajectories compared to baseline models. Another novel aspect is the use of full predictive distributions rather than point forecasts. Since diffusion models naturally generate multiple plausible futures, they capture the uncertainty inherent in free-living glucose behavior, something traditional models cannot express.

Finally, we observe that training over long horizons (up to 2000 epochs) allows the diffusion model to take advantage of the large number of overlapping windows per participant and learn stable personalized dynamics. Together, these choices make our approach meaningfully different from existing glucose forecasting methods and better aligned with the physiological patterns present in the data.

11 Team Contributions

Our team divided the work into major components while ensuring that everyone stayed involved in each stage of the project. Although each member had primary responsibilities, all of us contributed to discussions, debugging, literature review, experiment interpretation, and writing the reports.

11.1 Primary Responsibilities

Arunkumar and **Bright** led the preprocessing pipeline. They handled data cleaning, alignment of multimodal signals, sliding-window construction, and the creation of engineered features such as COB, rolling averages, and circadian encodings. They also performed quality checks to ensure that all participants’ data were consistent for modeling.

Ebrahim and **Prabhu** focused on adapting and training the diffusion model. They integrated our engineered features into the CSDI architecture, prepared conditional tensors, configured embedding layers, and managed the training loop and noise schedules. Ebrahim also mentored the group and helped resolve issues during model setup and experimentation.

Aman handled verification and validation. He evaluated model performance using RMSE, MAE, FID, and CRPS, compared diffusion models against baseline methods, and checked whether generated trajectories were physiologically realistic across participants.

11.2 Individual Achievements

Arunkumar strengthened his skills in multimodal preprocessing and feature engineering for physiological data. **Bright** gained experience in temporal data alignment and efficient window extraction. **Ebrahim** developed a deep understanding of diffusion models and guided the team through architectural and training challenges. **Prabhu** improved

in training generative models and tuning embedding strategies for multimodal inputs. **Aman** expanded his expertise in evaluation metrics and analysis of probabilistic forecasts.

11.3 Group Learning Outcome

As a group, we learned how to collaborate on a complex project that required coordination between data engineering, model development, and evaluation. Most importantly, every member now has a clear understanding of diffusion models and how conditional diffusion can be used for physiological time-series forecasting.

12 Conclusion

This project demonstrates that conditional diffusion models offer a strong and practical alternative to traditional glucose forecasting methods. By integrating multimodal physiological signals and engineered features, our adapted CSDI model learns meaningful patterns behind postprandial glucose changes and produces realistic future trajectories while also capturing uncertainty through full predictive distributions. Training over many epochs allowed the model to stabilize and consistently outperform baseline methods in capturing meal effects, circadian rhythms, and activity trends. Future extensions may include cross-participant generalization, incorporation of insulin and stress-related signals, and real-time deployment to further support personalized diabetes management tools.

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